Evaluate and **compare measured vs predicted MPI communication performance** across different cluster configurations.

Experiments to Perform

A. LITE-Pi Cluster

- Nodes: **6, 8, 10**
- Processes per node: **1,2 and 4**

B. CDAC Cluster

- Nodes: **2 nodes only**
- Cores: **32 cores and 48 cores**

Message Sizes

Run experiments using the following message sizes:

1 KB, 4 KB, 8 KB, 16 KB, 64 KB, 256 KB, 1 MB, 4 MB

Execution Guidelines

- Run experiments for **all mentioned combinations** of:
    - Nodes
    - Processes/Cores
    - Message sizes
- For physical experiments, perform **multiple runs (5–10)** and report **average execution time** (not needed for simulator runs)

Measured vs Predicted Analysis

You are required to:

- **Actually run the experiments** on both clusters (LITE-Pi and CDAC)
- Collect **measured execution time**

- Generate **predicted results** using your model/simulation (e.g., LogGP / SMPI / analytical model)
- **Compare measured vs predicted performance**

Output Format

Prepare **TWO separate CSV files, one for each cluster**

Each CSV must include:

- Number of Nodes
- Processes/Cores
- Message Size
- Measured Execution Time
- Predicted Execution Time (SimGrid)
- Predicted Execution Time (LogGOPSim)

Make comparative plots (time vs message size) for each configuration (nodes and processes-per-node).

Three figures are expected in each plot: Measured, SimGrid prediction, and LogGOPSim prediction.